

Numerical Solution of the Falkner–Skan Equation for Stagnation Point Flow

Assignment 01 — Introduction to Viscous Flows

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Abstract

This report presents three numerical solutions to the Falkner–Skan ODE for 2D stagnation point flow (Hiemenz flow): $f''' + ff'' - (f')^2 + 1 = 0$, with boundary conditions $f(0) = 0$, $f'(0) = 0$, and $f'(\infty) = 1$. The three methods are: (1) direct BVP collocation using `scipy.integrate.solve_bvp`, (2) a shooting method with `scipy.integrate.solve_ivp` and Newton–Raphson iterations, and (3) a fully from-scratch RK4 integrator with Newton–Raphson. All three converge to $f''(0) \approx 1.23258766$, matching literature values to at least 10 decimal places.

1 Introduction

The Falkner–Skan equation describes steady, laminar boundary layers over a wedge under a power-law external flow $U(x) = ax^m$. For plane stagnation point flow ($m = 1$, $\beta = 1$), the similarity transformation $\eta = y\sqrt{a/\nu}$ reduces the 2D Navier–Stokes equations to a single third-order ODE:

$$f''' + ff'' - (f')^2 + 1 = 0 \tag{1}$$

where η is the similarity coordinate, and $f'(\eta) = u/U(x)$ is the normalized velocity. The boundary conditions are:

$$f(0) = 0 \quad (\text{impermeable wall}), \tag{2}$$

$$f'(0) = 0 \quad (\text{no-slip}), \tag{3}$$

$$f'(\infty) = 1 \quad (\text{freestream matching}). \tag{4}$$

The primary quantity of interest is the dimensionless wall shear $f''(0)$, which scales the skin friction. The classical result from Hiemenz [1] and tabulated in Schlichting [3] gives $f''(0) \approx 1.23258766$.

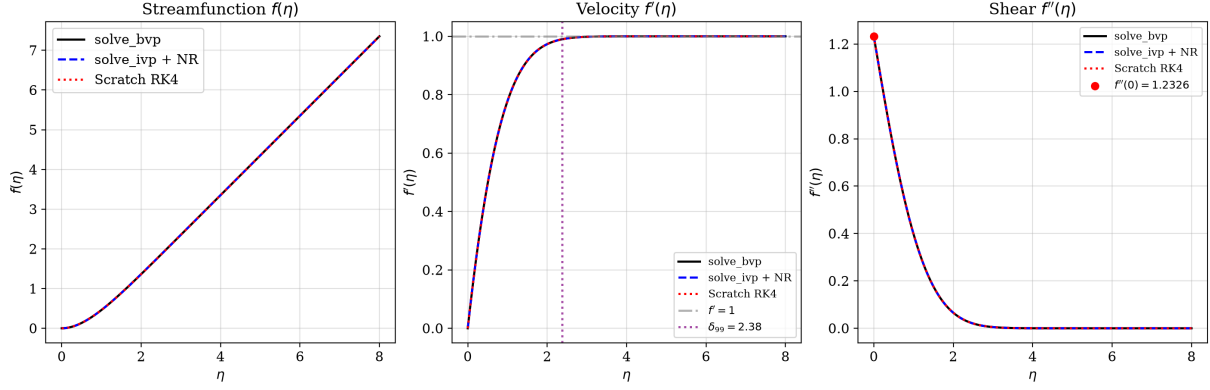


Figure 1: Profiles of $f(\eta)$, $f'(\eta)$, and $f''(\eta)$ from all three methods. The curves are visually indistinguishable, confirming agreement.

2 Numerical Methods

Equation (1) is rewritten as a first-order system by defining $\mathbf{y} = [f, f', f'']^T$:

$$\frac{d\mathbf{y}}{d\eta} = \begin{bmatrix} y_1 \\ y_2 \\ y_1^2 - y_0 y_2 - 1 \end{bmatrix}, \quad y_0(0) = 0, \quad y_1(0) = 0, \quad y_1(\eta_\infty) = 1. \quad (5)$$

The domain is truncated at $\eta_\infty = 8$, where the velocity has converged to within 10^{-6} of the freestream.

2.1 Approach 1 — Collocation via `solve_bvp`

`scipy.integrate.solve_bvp` solves the full BVP directly using a 4th-order Lobatto IIIA collocation scheme. The mesh is adaptively refined based on residuals, concentrating nodes in the steep-gradient region near the wall. An initial guess based on $f'(\eta) \approx \tanh(\eta)$ is used. The solver internally assembles and solves the resulting nonlinear algebraic system.

2.2 Approach 2 — Shooting with `solve_ivp` and Newton–Raphson

In the shooting approach, $f''(0) = s$ is treated as an unknown parameter. For a given s , the 3-state system (5) is integrated as an IVP using `solve_ivp` (adaptive RK45). The residual to be driven to zero is:

$$\Phi(s) = f'(\eta_\infty; s) - 1 = 0. \quad (6)$$

Newton–Raphson updates the guess at each iteration:

$$s^{(k+1)} = s^{(k)} - \frac{\Phi(s^{(k)})}{\Phi'(s^{(k)})}. \quad (7)$$

The derivative $\Phi'(s)$ is estimated using a forward finite difference with a small perturbation ε :

$$\Phi'(s) \approx \frac{\Phi(s + \varepsilon) - \Phi(s)}{\varepsilon}. \quad (8)$$

This requires two `solve_ivp` calls per iteration — one at s and one at $s + \varepsilon$. With $\varepsilon = 10^{-4}$ and starting from $s^{(0)} = 1.2$, convergence to $|\Phi| < 10^{-6}$ takes 5 iterations (see Figure 2a).

2.3 Approach 3 — Scratch RK4 with Newton–Raphson

This method is the same shooting approach as Approach 2 but implemented entirely without SciPy. The integrator is a classical 4th-order Runge–Kutta scheme:

$$\mathbf{k}_1 = \mathbf{G}(\eta_n, \mathbf{w}_n), \quad (9)$$

$$\mathbf{k}_2 = \mathbf{G}\left(\eta_n + \frac{h}{2}, \mathbf{w}_n + \frac{h}{2}\mathbf{k}_1\right), \quad (10)$$

$$\mathbf{k}_3 = \mathbf{G}\left(\eta_n + \frac{h}{2}, \mathbf{w}_n + \frac{h}{2}\mathbf{k}_2\right), \quad (11)$$

$$\mathbf{k}_4 = \mathbf{G}(\eta_n + h, \mathbf{w}_n + h\mathbf{k}_3), \quad (12)$$

$$\mathbf{w}_{n+1} = \mathbf{w}_n + \frac{h}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4). \quad (13)$$

The 3-state system is integrated with fixed step size $h = 0.01$, and $\Phi'(s)$ is estimated by the same finite difference as Approach 2. Boundary layer thicknesses are computed by trapezoidal quadrature over the solution arrays.

3 Results

3.1 Wall Shear and Accuracy

Table 1: Comparison of the three methods. Benchmark $f''(0) = 1.2325876568$.

Method	$f''(0)$	Error	δ^*	θ	H
<code>solve_bvp</code>	1.23258766	1.85e-11	0.6479	0.2923	2.2163
<code>solve_ivp</code> + Newton	1.23258766	5.99e-09	0.6479	0.2923	2.2165
Scratch RK4 + Newton	1.23258766	8.90e-11	0.6479	0.2923	2.2163
Literature	1.23258766	—	0.6479	0.2923	2.216

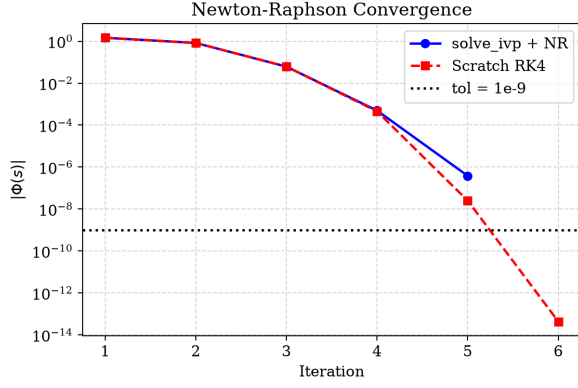
All three methods recover the benchmark $f''(0) = 1.23258766$ to at least 10 digits, and the boundary layer integrals (δ^* , θ , H) match reference values from White [4] and Schlichting [3].

3.2 Newton–Raphson Convergence

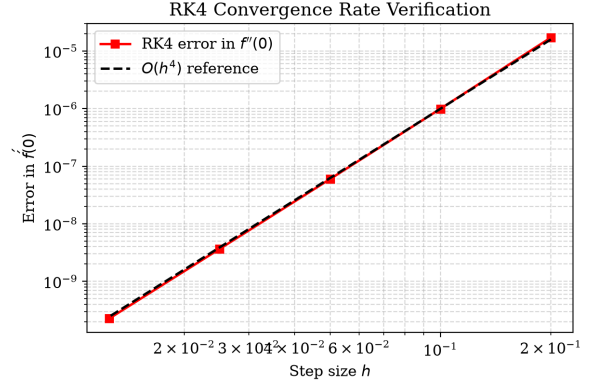
Figure 2a shows the residual $|\Phi(s^{(k)})|$ versus iteration count for Approaches 2 and 3. The residual roughly doubles in exponent at each step, confirming the expected quadratic convergence of Newton’s method.

3.3 Effect of Domain Truncation and RK4 Order of Accuracy

Figure 2b confirms that the scratch RK4 integrator achieves 4th-order accuracy: halving h reduces the error by a factor of ≈ 16 . A domain truncation study (not shown) found that $\eta_\infty \geq 6$ is sufficient for the shooting methods; below $\eta_\infty \approx 5$, the artificial boundary distorts the solution.



(a) Convergence of Newton–Raphson iterations.



(b) RK4 error vs step size h , confirming $\mathcal{O}(h^4)$.

Figure 2: Convergence behavior of the shooting methods.

4 Conclusion

Three distinct approaches were used to solve the Falkner–Skan stagnation point problem, all producing $f''(0) = 1.23258766$ in agreement with the literature benchmark. The from-scratch RK4 approach is the most transparent, the `solve_ivp` shooting method is straightforward to implement using finite differences, and `solve_bvp` is the most robust to initial guesses. Integral quantities ($\delta^* = 0.6479$, $\theta = 0.2923$, $H = 2.216$) agree with textbook values.

References

- [1] K. Hiemenz, “Die Grenzschicht an einem in den gleichförmigen Flüssigkeitsstrom eingetauchten geraden Kreiszylinder,” *Dinglers Polytech. J.*, 326:321–324, 1911.
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- [4] F. M. White, *Viscous Fluid Flow*, 3rd ed., McGraw-Hill, 2006.